PACKET 1 • SOFTWARE DEVELOPMENT

4 Make AI Challenge Its Own Work

RefView Window/Level Display Transform

Review stance Assume your current solution contains mistakes. Use AI as an independent reviewer, not as the original author.

Ask the reviewer to look for

- Incorrect assumptions
- Off-by-one errors
- Boundary-condition errors
- Rounding errors
- Missing tests
- Invalid-input failures
- Differences between the implementation and the specification

Review requirements

- Provide a concrete counterexample for each suspected defect wherever possible.
- Identify the exact specification or acceptance criterion involved.
- Distinguish confirmed defects from questions or stylistic preferences.
- Recommend the smallest change that resolves each confirmed defect.

Team decision

- Decide which findings are valid.
- Update the solution and tests.
- Keep a brief record of one review finding you rejected and why.

Success check Can the final tests demonstrate that the reviewer's valid counterexamples are now handled correctly?

PACKET 2 • RESEARCH SCIENTISTS

4 Make AI Challenge Its Own Work

RefView SmoothCT Experimental Evaluation

Review stance Assume the current plan has weaknesses. Ask AI to act as a skeptical scientific reviewer.

Have the reviewer examine

- Confounding
- Independence of observations
- Measurement bias
- Selection bias
- Insufficient replication
- Inappropriate statistical analysis
- Uncertainty
- Whether the experiment tests the complete hypothesis

Review requirements

- Explain why each weakness matters.
- Describe the incorrect conclusion it could produce.
- Identify the affected acceptance criterion or threshold.
- Propose a concrete design or analysis improvement.
- Separate fatal validity threats from useful refinements.

Team decision

- Decide which criticisms are valid.
- Improve the plan.
- Record the most consequential weakness discovered.

Success check Does the revised experiment test noise, resolution, and CT-number bias together rather than treating one favorable metric as sufficient?

PACKET 3 • PRODUCT DEVELOPMENT

4 Make AI Challenge Its Own Work

RefView Auto-Window Capability

Review stance Assume the current requirements and verification strategy contain important gaps. Use an independent system/design reviewer.

Have the reviewer challenge

- Ambiguous requirements
- Requirements that cannot be objectively verified
- Missing failure behavior
- Missing edge cases
- Technical risks
- Verification gaps
- Assumptions about image content
- Traceability between requirements, risks, and tests

Review requirements

- Identify specific failure scenarios, not only general concerns.
- Explain the consequence of each failure.
- Identify the affected requirement or acceptance criterion.
- Propose measurable wording, mitigation, or verification evidence.
- Flag requirements that use subjective terms without objective criteria.

Team decision

- Decide which findings are valid.
- Improve the requirements, risks, and tests.
- Record one major gap that was not visible in the initial draft.

Success check Can every important behavior and risk be connected to objective evidence and a clear pass/fail decision?